

Physics Question Bank
Questions
Unit 10

Knowledge covered:

الوحدات التدريبية	Unit 10: Electromagnetic induction	
أكواد المعارف	TPK03 – TPK04 - TPK05	
المعارف	TPK03	Meaning of Electromagnetic Induction and Faraday's Law
	TPK04	Lenz's Law
	TPK05	Mutual and self-induction The electric generator (dynamo) The electric transformer

Knowledge in each lesson:

Unit 10	Lesson 1	Lesson 2	Lesson 3	Lesson 4	Lesson 5
Knowledge code	TPK03	TPK04	TPK05	TPK05	TPK05

Questions expected in almost each lesson:

1. Write the scientific term expressed by the following statements.
2. Mention the factors affecting each of the following (one factor only)
3. Put circle around the correct answer
4. Correct the underlined in the following statement
5. True or False
6. Answer the following question
 - i. Solve questions (calculate / find / determine)
 - ii. Mention the rule or method used to determine
 - iii. Draw the shape
 - iv. State
 - v. Write the mathematical law used to calculate
 - vi. Give a reason for each of the following
7. Compare between each of the following

Lesson (1): Meaning of electromagnetic induction and faraday's law

1. Write the scientific term expressed by the following statements.

No.	Question	Answer
1	phenomenon in which an induced electromotive force and an induced current are generated in the conductor by a changing magnetic field (magnetic flux). (دور أول). (دور ثان ٢٠٢٣) (دور ٢٠٢٤)	
2	An electromotive force (current) is induced in a conductor while it is moving (cutting) across the magnetic field.	

2. Mention the factors affecting each of the following (one factor only)

No.	Question	Answer
1	the direction of the induced emf by electromagnetic induction.	
2	emf induced by electromagnetic induction (faraday's experiment).	

3. Put circle around the correct answer

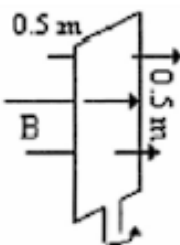
No.	Question	Choices	Answer
1	A moving conductor coil produces an induced e.m.f. This is in accordance with	a) Lenz's law b) Faraday's law c) Coulomb's law	
2	Weber per square meter is a) henry b) ampere c) Tesla d) Ohm		
3	Which of the following statements is true about an electromagnet? a) An electromagnet is permanent magnet b) The strength of an electromagnet can be changed by changing the number of turns in its coil c) An electromagnet produces a comparatively weak force of attraction d) The polarity of an electromagnet cannot be changed		

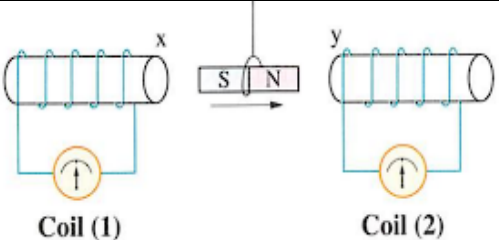
4	If a coil is placed in a uniform magnetic field, then a) No induced emf generated. b) an induced current flow in the coil. c) an emf is induced in the coil. d) None of the above	
5	A coil of copper wire is moved parallel to a uniform magnetic field. The e.m.f. induced in the coil will: (دور أول ٢٠٢٥) a) depend on the intensity of magnetic field b) depend on the velocity of the coil c) be zero d) be infinite	
6	In Faraday's Experiment for electromagnetic induction, the induced emf in the coil increases when: (تجريبي ٢٠٢٥) a) Keeping the magnet still inside the coil. b) Speeding up the magnet motion relative to the coil. c) Connecting a galvanometer to the coil. d) Increasing the spacing between the coil turns.	

4. Correct the underlined in the following statement

No.	Question	Answer
1	The process of producing electric current through the variable magnetic field is called <u>Faraday's Law</u>	
2	According to Faraday's law of electromagnetic induction, electromotive force is produced by time varying magnetic <u>force</u> .	
3	If the speed by which a magnet approaching a solenoid is <u>constant</u> , then the induced e.m.f. will be zero	
4	The polarity of induced e.m.f. is given by <u>Faraday's law</u>	
5	The magnitude of the induced e.m.f. in a conductor depends on the <u>speed</u> of change of the magnetic flux within the conductor	
6	If the number of turns of the metal coil cutting the variable magnetic flux increases at a constant rate, then the average induced electromotive for <u>decreases</u> .	

5. Answer the following question

N no	Question	Answer
1	A loop of an area 2 m^2 is placed in a magnetic field which changes from 4 Wb/m^2 to 8 Wb/m^2 in 2 seconds. Find the induced e.m.f. in the coil	
2	A square loop is placed in a uniform magnetic field perpendicular to the plane of the loop as shown. The loop is 0.50 meters on a side and the magnetic field B has a strength of 2 T. If the loop is rotated through an angle of 90° in 0.1 second what would be the average induced EMF in the loop? 	
3	A square coil having 500 loops of side 10 cm is placed normal to magnetic flux which increases at a rate of 1 T/s . The induced emf is	
4	following figure shows a magnet placed between two coils, each of them is connected to a moving coil galvanometer whose zero reading is at the middle of its scale. If the magnet is moved in the shown direction as in the figure:	

	 Coil (1) Coil (2) a) Determine on the drawing the direction of the induced current generated in the two coils. b) Determine the type of the magnetic poles which are formed at x. y.	
5	A coil of 200 turns intercepts magnetic flux of 7×10^{-3} Wb. If this magnetic flux density has changed uniformly to 5×10^{-3} Wb within 0.1 s, calculate the induced electromotive force in the coil	
6	A square coil of side length 10 cm and 500 turns is placed perpendicular to a uniform magnetic field of flux density 0.1 T. Calculate the induced emf in the coil, if a) The magnetic flux fades within 0.15 s. b) The flux density is increased to 0.3 T within 0.75 s	
7	The magnetic flux through a 50 turns coil increases at the rate of 0.05 Wb/s. What is the induced emf between the ends of the coil?	

8	The magnetic flux in a coil change from 20×10^{-2} Weber to 12×10^{-2} Weber in 0.02 s. Find the value of induced emf in the coil.	
9	A coil of an area 0.2 m^2 is placed in a magnetic field which changes from 5 Wb/m^2 to 10 Wb/m^2 in 0.2 seconds. Find the induced e.m.f. in the coil	
10	Calculate the magnitude of the induced emf when the magnet is thrust into the coil, given the following information: the single loop coil has a radius of 6.00 cm and the average value of $B \cos\theta$ (this is given, since the bar magnet's field is complex) increases from 0.0500 T to 0.250 T in 0.100	
11	A coil having 1000 loops of side length 20 cm is placed normal to a magnetic flux which increases at a rate of 0.5 T/s. then induced e.m.f. is	
12	A coil of area 50 cm^2 has 1000 turns. Magnetic field of 0.2 weber/m^2 is perpendicular to the coil. The magnetic field is reduced to zero in 0.2 second. The induced emf in the coil is	
13	A coil of 800 turns, each of area 36 cm^2 , the coil is affected by a normal magnetic flux relative to its plane,	

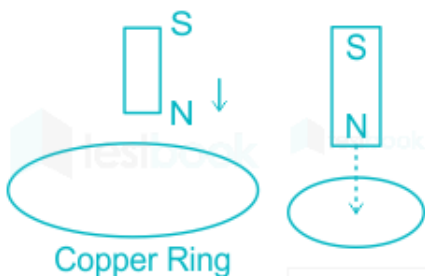
	the flux density is $5.3 \times 10^{-6} \text{T}$, if the magnetic flux is vanished in 1.5 ms. Calculate the induced EMF.	
14	Mention the application of electromagnetic induction.	
15	Mention the scientific base electric generators.	
16	A square loop of 10 turns is placed in a uniform magnetic field perpendicular to the plane of the loop, if the magnetic flux changes from 0.03 Wb to zero in 0.1 s. Calculate the average induced emf in the loop. (تجريبي)	
17	A metallic ring is placed in a uniform magnetic field perpendicular to the plane of the ring. If the magnetic flux changes from 0.05 Wb to -0.05 in 0.05 s. Calculate the average induced EMF in the loop. (دور أول 2023)	
18	Mention the mathematical formula for average induced emf in a coil. (تجريبي ٢٠٢٥)	

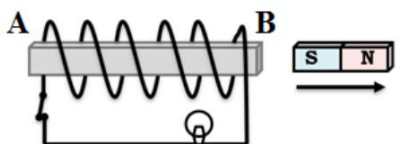
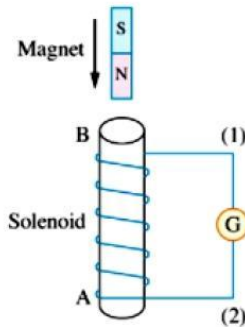
Lesson (2): Lenz's law

1. Write the scientific term expressed by the following statements.

No.	Question	Answer
1	The negative sign in faraday's relation that the direction of the induced electromotive force or the induced current tends to oppose the cause producing it.	
2	The induced current must be in a direction such as to oppose the change producing it. (تجريبي)	

2. Put circle around the correct answer

No.	Question	Choices	Answer
1	The induced current must be in a direction such as to oppose the change producing it	a) Electromagnetic Induction b) Lenz's law c) Faraday's Law	
2	A bar magnet is dropped and it passes through a copper ring as shown in the diagram. The acceleration of the falling magnet while passing through the ring is 	a) equal to acceleration due to gravity b) more than the acceleration due to gravity c) less than the acceleration due to gravity d) Zero	
3	Identify the law used to find the direction of eddy current	a) Lenz's law b) Faraday's Law	
4	When a conductor connected to an electrical circuit moves in a magnetic field, the direction induced current depends upon	a) number of turns and length of the conductor b) strength of the magnetic field c) speed of movement of the conductor d) direction of the magnetic field and direction of motion of the conductor	

5	When the magnet is plugged out of the coil, the induced current in the coil will experience	a) Attraction force b) Repulsion force c) Need more givens																
6	When the magnet is plugged into the coil, the induced current in the coil will experience	a) Attraction force b) Repulsion force c) Need more givens																
7	A permanent magnet is quickly introduced into the center of an aluminum closed ring. With this ring ...	a) will be attracted to the magnet b) push away from the magnet c) remain stationary																
8	In the opposite figure, when the magnet moves in the direction shown the pole formed at A is ... (دور ثانی 2023) 	a) North b) South c) West d) east																
9	A bar magnet falls towards a solenoid as in the opposite figure, then during the approaching of the magnet towards the coil, which of the following choices? <table border="1" data-bbox="183 1115 971 1434"><thead><tr><th></th><th>The direction of the current in the galvanometer</th><th>The type of the formed pole at (A)</th></tr></thead><tbody><tr><td>(A)</td><td>From (2) to (1)</td><td>north</td></tr><tr><td>(B)</td><td>From (1) to (2)</td><td>south</td></tr><tr><td>(C)</td><td>From (2) to (1)</td><td>south</td></tr><tr><td>(D)</td><td>From (1) to (2)</td><td>north</td></tr></tbody></table>		The direction of the current in the galvanometer	The type of the formed pole at (A)	(A)	From (2) to (1)	north	(B)	From (1) to (2)	south	(C)	From (2) to (1)	south	(D)	From (1) to (2)	north		
	The direction of the current in the galvanometer	The type of the formed pole at (A)																
(A)	From (2) to (1)	north																
(B)	From (1) to (2)	south																
(C)	From (2) to (1)	south																
(D)	From (1) to (2)	north																
10	According to the Lenz's law..... (دور أول 2024) a) The direction of the induced emf will be such that it helps the change in magnetic flux b) The direction of the induced current will be such that it opposes the change in magnetic flux. c) The magnitude of the induced electromotive force (emf) is directly proportional to the rate of change of flux. d) The direction of the induced electromotive force (emf) will be along the direction of the magnetic field.																	

3. Correct the underlined in the following statement

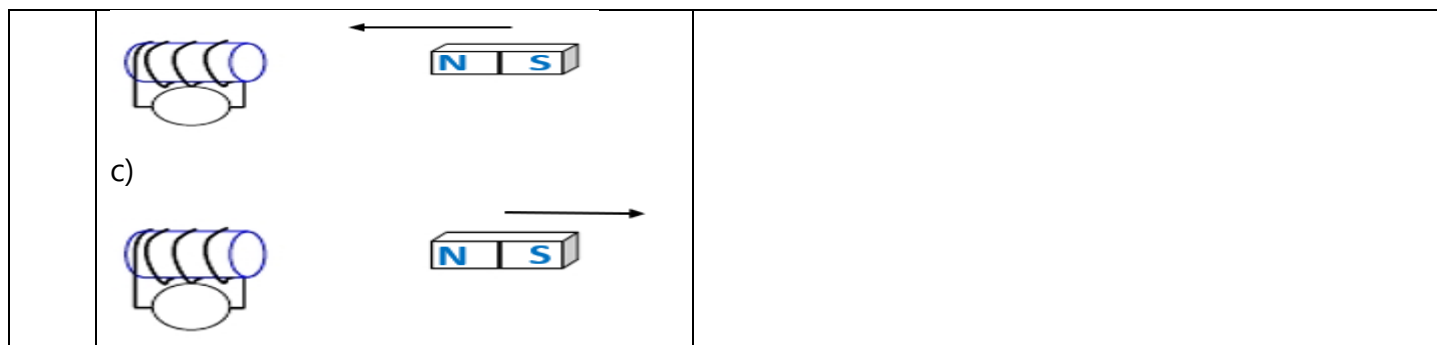
No.	Question	Answer
1	The coefficient of mutual inductance between two coils increases as the rate of change of current in the primary coil decreases. (2023 دور ثان)	
2	Lenz's law is used to determine the magnitude of emf induced in the coil	

4. True or False

No.	Question	Answer
1	Lenz's law is used to determine the magnitude of emf induced in the coil	
2	Induced current will appear in such a direction that it opposes the change that produced it	
3	The Lenz's law is consistent with the law of conservation of energy	
4	When the current flowing through a wire reverses direction, the magnetic field around the wire reverse direction	
5	If the current in a wire double, the induced magnetic field remain the same	
6	Magnetic flux is maximum when the plane of the loop is parallel to the magnetic field	
7	From Lenz's rule, the direction of induced current in generated in the conductor depends on the direction of the motion and the direction of the magnetic field. (تجريبي ٢٠٢٥)	
8	When the magnet is moved away from the solenoid, magnetic flux link with the coil increases, the coils near end forms like pole which tends to repel it. (تجريبي ٢٠٢٥)	

5. Answer the following question

No.	Question	Answer
1	In the opposite figure: What is the direction of the induced EMF? a)  b)	



6. Compare between each of the following

No.	Question		Answer
1	Compare (دور أول 2023)		
	Point of comparison	Lenz's rule	The right cork screw rule
	The use		
2	Compare (تجريبي ٢٠٢٥)		
	Point of comparison	Fleming's right hand rule	Lenz's rule
	Uses		

Lesson (3): Mutual and self-induction

1. Write the scientific term expressed by the following statements.

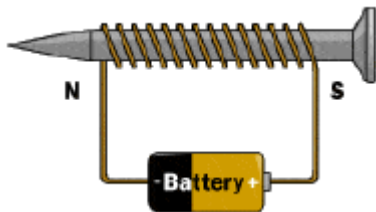
No.	Question	Answer
1	It's the electromagnetic effect takes place between two coils when an induced emf generated in one of them (secondary coil) due current variation in the other coil (primary coil), which opposes the change causing it. (تجريبي)	
2	The phenomenon of inducing emf in a coil due to change in current in the same coil and hence the change in magnetic flux in the coil. (تجريبي ٢٠٢٥)(2023 دور أول)	

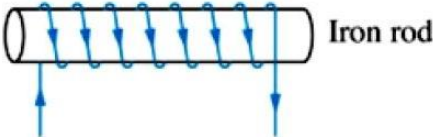
2. Mention the factors affecting each of the following (one factor only)

No.	Question	Answer
1	emf induced by mutual induction	
2	emf induced by self-induction	
3	The coefficient of mutual inductance between two coils. (2024 دور أول ٢٠٢٣) (دور أول)	
4	The self-inductance (coefficient of self-induction)	

3. Put circle around the correct answer

No.	Question	Choices	Answer
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1	A transformer is based on the principle of	a) Mutual-inductance b) Self-inductance c) Both a and b	
2	Changing current in a coil produces emf in the same coil. this phenomenon is known as	a) Mutual-inductance b) Self-inductance c) Both a and b	
3	The phenomenon due to which there is an induced current in one coil due to the current in a neighboring coil is?	a) Mutual-inductance b) Self-inductance c) Both a and b	
4	If you wrap more coils around an electromagnet, it will become... 	a) Brighter b) Weaker c) Invisible d) Stronger	
5	The production of electric current by a changing magnetic field is called	a) induced current b) electromotive force c) electromagnetic induction	
6	The effect on inductance by adding extra coil layers	a) increase L b) decrease L c) Nothing happens d) Need more givens	
7	The production of a voltage across an inductor as a result of its own expanding or contracting magnetic field is called	a) Mutual-inductance b) Self-inductance c) Both a and b	
8	Henery is the unit of the (دور أول ٢٠٢٥) a) Coefficient of mutual inductance only b) Coefficient of self-inductance only c) Both a and b d) None		
9	When do we have self-induction? a) When the magnet moves into the coil b) When the coil moves into the magnet c) When a constant current passes through the coil d) When a changing current passes through the coil		
10	A coil which carries current I is wound on iron core. The self-induced electromotive force in the coil is not affected by..... (دور أول 2024)		

	a) Variation in coil current b) Variation in voltage to the coil c) Change of number of turns of coil d) The resistance of coil wire	
11	In the opposite figure, when the iron rod is withdrawn from the coil, the self inductance of the coil (A) becomes zero. (B) decreases. (C) increases. (D) remains constant. 	
12	When the electric intensity passing through a coil of self-inductance 0.014H changes from 10A to zero, an induced emf of 7V is generated between the terminals of the coil. So, the time of the change in the electric current intensity is (A) 0.02s (B) 0.01s (C) 0.04s (D) 0.03s	
13	The coil of self-induction 0.05H consisting of 100turns carries an electric current that produces a magnetic flux of $9 \times 10^{-4} \text{Wb}$ through the coil. If the passing current in the coil has vanished through an interval of 0.03s, then the average induced emf in the coil equals and the intensity of the current which was flowing in the coil before it vanishes equals (A) 3V, 3.6A (B) 3V, 1.8A (C) 12V, 3.6A (D) 12V, 1.8A	

14	A solenoid of 500 turns has a self-inductance of 0.5H. If the intensity of the current that passes through the solenoid changes by (ΔI) in amperes, then the value of the change in the magnetic flux that cuts the solenoid during the same time is numerically equal to (A) $(0.01\Delta I)Wb$ (B) $(10\Delta I)Wb$ (C) $(50\Delta I)Wb$ (D) $(0.001\Delta I)Wb$	
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4. Correct the underlined in the following statement

No.	Question	Answer
1	The induced electromotive force (emf) generated in a coil increases to <u>double</u> if the rate of change of current in the same coil increases 4 times. (دور أول 2024)	
2	The mutual induction coefficient measured in <u>Ampere/second</u> . (دور أول 2023)	
3	There is an induced emf in the secondary coil only when the current in the primary coil is <u>constant</u> . (تجريبي ٢٠٢٥)	
4	Switching on the current in primary coil or increasing the current intensity in primary coil by using rheostat producing <u>forward</u> induced emf. (تجريبي ٢٠٢٥)	

5. True or False

No.	Question	Answer
1	Switching the current in the primary coil on and off will produce an induced electromotive force in the secondary coil. (دور أول ٢٠٢٥)	
2	The mutual-inductance between two coil depends on number of turns in each of the two coils.	
3	The mutual-inductance between two coil depends on the medium inside the coils	
4	The self-inductance of a coil depends on the medium inside the coil.	

5	The self-inductance of a coil depends on the current flowing through the turns of the coil.	
6	The self-inductance of a coil depends on the separation between the turns of the coil	
7	Self-induced emf is always proportional to the time rate of change of the current.	
8	When a constant current of 5.0 A flows in a 400 turns solenoid of length, l and cross-sectional area A , emf is induced in the coil.	
9	Self-inductance is vector	
10	the voltage induced across an inductor is inversely proportional to the rate of change of magnetic flux.	
11	When the rate of change in the current intensity in a coil increases into double, the self-inductance coefficient of the coil doubled. (دور أول ٢٠٢٥)	

6. Answer the following question

No.	Question	Answer
1	When current in a coil change from 5 A to 2 A in 0.1 s, average voltage of 50 V is produced. The self-inductance of the coil is	
2	Determine the inductance of the coil if a -5 V self-induction EMF occurs at a rate of change of current of 10 A / s.	
3	When the current in a coil is reduced uniformly from 20 A to zero in 0.50 s, an e.m.f. of 4.0 V is induced in a neighboring coil. What is the value of the mutual inductance between the two coils?	
4	An application of mutual induction is	

5	An application of self-induction	
6	When current in a coil change from 6 A to 1 A in 0.2 seconds, an average voltage of 40 V is produced. The self-inductance of the coil is?	
7	When a current of 2mA is supplied to a coil with 100 turns, a magnetic flux of magnitude 0.2Wb is linked with it. Find the self-inductance of this coil.	
8	What is the principle of mutual induction?	
9	A long solenoid has 500 turns. When a current of 2 A is passed through it, the resulting magnetic flux linked with each turn of the solenoid is 4×10^{-3} Wb. find self-inductance.	
10	When a current of 4A between two coils changes to 12A in 0.5s in primary and induces an emf of - 50mV in the secondary. Calculate the mutual inductance between the two coils.	
11	Mention the application of 1- mutual induction. 2- self-induction.	

12	Mention the scientific base 1- electric transformer. 2- neon lamp.	
13	An electric current of intensity 5 A flowing through a coil of self-inductance 0.001 H. If the current vanishes in 0.5 second, calculate the induced electromotive force (emf) generated in the coil. (دور ثان) (2023)	
14	when the current changes from +2A to - 2A in 0.05 sec, an emf induced in a coil 8V Calculate the coefficient of self-induction in the coil. (تجريبي)	
15	When the current in a coil change from 5A to 2A in 0.05 s, an e m f of 15 V is induced in the coil. Calculate the coefficient of self-induction of the coil. (دور أول) (2023)	
16	Mention the mathematical formula for mutual induction coefficient. (تجريبي ٢٠٢٥)	

7. Compare between each of the following

No.	Question		Answer
1	Compare		
	Point of comparison	Backward induced EMF	Forward induced EMF
	Movement		
	Using rheostat		

	Using switch			
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Lesson (4): The electric generator (dynamo)

1. Write the scientific term expressed by the following statements.

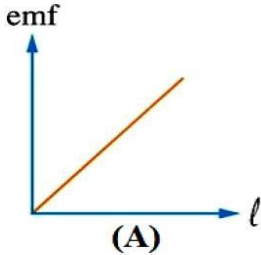
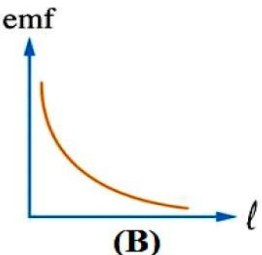
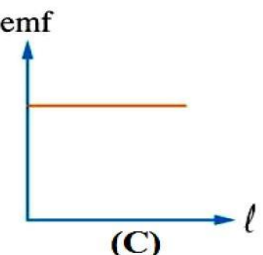
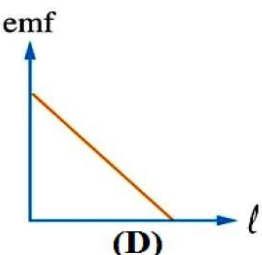
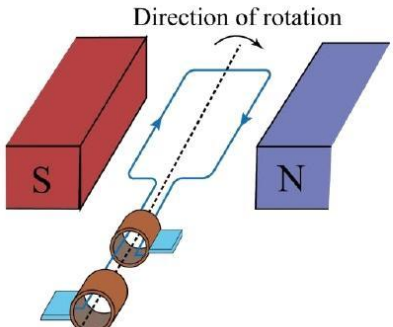
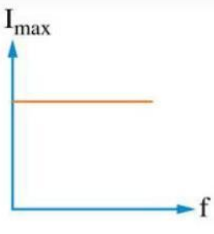
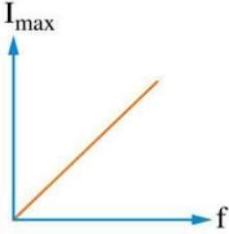
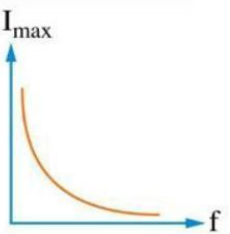
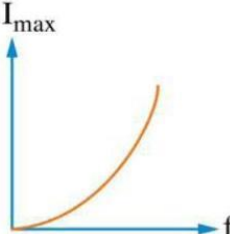
No.	Question	Answer
1	It is a device used for converting mechanical energy into electrical energy. (دور أول ٢٠٢٥)	

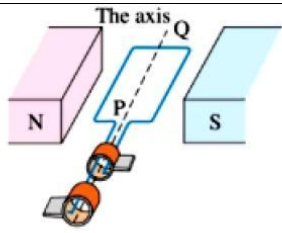
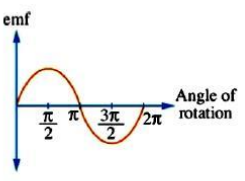
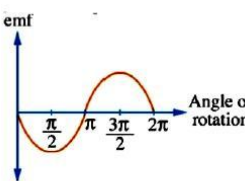
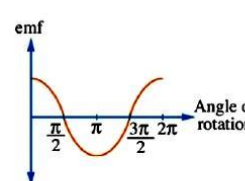
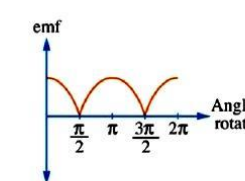
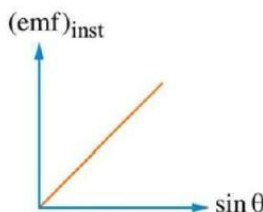
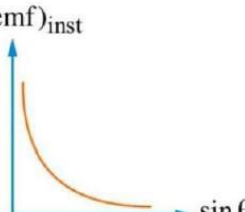
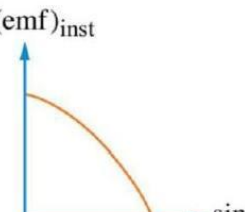
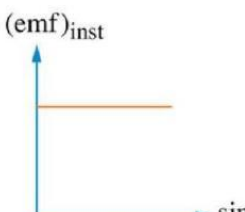
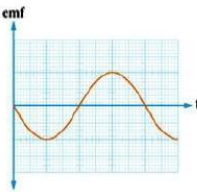
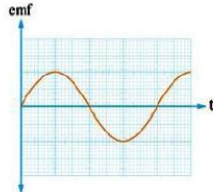
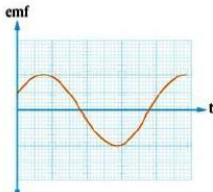
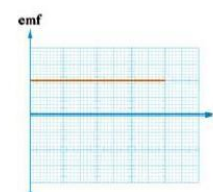
2. Mention the factors affecting each of the following (one factor only)

No.	Question	Answer
1	The induced emf in a straight wire moving normal to magnetic field. (تجريبي) (دور أول ٢٠٢٣)	
2	The direction of induced emf in a straight wire moving normal to a magnetic field	
3	The instantaneous value of induced emf generated in the dynamo. (دور أول 2024)	

3. Put circle around the correct answer

No.	Question	Choices	Answer
1	When a coil rotates in a magnetic field, the direction of the induced electromotive force produced in the coil is changed eachcycle (دور ثان 2023)	a) 1/4 b) 1/2 c) 1/1 d) 3/4	
2	When the number of turns in the dynamo's coil is doubled at constant other factors then the maximum electromotive force generated from it..... (دور أول ٢٠٢٣) (دور ثان 2023)	a) Increases to the double b) Decreases to half of its value c) remains constant d) Increases three times of its value	

3	Which of the following graphs represents the relation between the induced electromotive force (emf) between the terminals of a wire that has a changeable length and the length of the wire (L), when it moves each time with the same uniform velocity perpendicular to a uniform magnetic field?  (A)  (B)  (C)  (D)	
4	The opposite figure represents an AC dynamo, if the output voltage was 10V when the dynamo's position was as in the figure, so it becomes (-10V) when the coil rotates by an angle of (A) 270° (B) 180° (C) 90° (D) 360°  Answer:	
5	Which of the following graphs represents the relation between the maximum value of the AC current ($I_{\max}$) which is produced from a dynamo connected to an ohmic resistance and the frequency of the rotation of the dynamo's coil (f)?  (A)  (B)  (C)  (D)	

6	The opposite figure shows a rectangular coil that rotates between the poles of a magnet. If the coil rotates about its axis (PQ) starting from the position shown in the opposite figure, so which of the following graphs represents the change of the induced emf in the coil through one complete cycle?   (A)  (B)  (C)  (D)	
7	Which of the following graphs represents the relation between the magnitude of the instantaneous induced electromotive force $(emf)_{inst}$ in the dynamo's coil and the sine of the angle of rotation of the coil $(\sin \theta)$ starting from zero position?  (A)  (B)  (C)  (D)	
8	which of the following graphs represents the relation between the induced instantaneous electromotive force and time in a simple dynamo, if the coil started rotating from the position where the plane of the coil was inclined to the field by an angle of 60°?  (A)  (B)  (C)  (D)	

9	The magnitude of the instantaneous induced electromotive force which is generated in the dynamo's coil when the magnetic flux that penetrates it is the maximum value equals (A) the effective value. (B) zero. (C) the maximum value. (D) the average value.	
10	which of the following will cause an induced current in a coil of wire (تجريبي) a) A magnet resting near the coil. b) the constant field of the earth passing through the coil c) A magnet being moved into or out of the coil. d) A wire carrying a constant current near the coil.	
11	A straight wire 1m long is moving in a uniform magnetic field of flux density 0.1T at a uniform velocity 10m/s. an induced potential difference of 0.5V is generated between its terminals. The angle between the direction of the wire motion and the direction of the magnetic field is. (2024 دور أول) a) 0° b) 30° c) 60° d) 90°	
12	A simple AC generator is made by rotating a flat rectangular coil in a uniform magnetic field. The instantaneous value of emf produced when the coil is parallel to the magnetic field is (100 V). If the frequency of rotation is doubled, then the instantaneous value of emf produced at the same position is..... (2023 دور أول) (a) 50V (b) 100V (c) 200V (d) 400V	
13	When the number of turns in the dynamo's coil is doubled, then the maximum generated electromotive force generated from it: (٢٠٢٥ تجريبي) a) Increases to double b) Decreases to its half value c) Remained constant d) Increases 4 times	
14	A rectangular coil of 100 turns and cross-sectional area 0.06 m ² rotates at frequency 50 Hz in a uniform magnetic field of flux density 0.1 T. The maximum induced emf is (تجريبي ٢٠٢٥) (A) 30 V (B) 94.2 V (C) 188.5 V	

(D)	60 V	
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4. Correct the underlined in the following statement

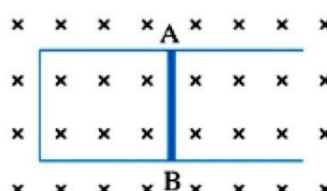
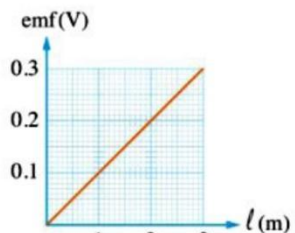
No.	Question	Answer
1	The ac generator is a device based on the principle of <u>magnetic torque</u> . (دور أول 2024)	
2	The electromotive force induced in a rectangular coil rotating in a magnetic field becomes zero when the coils' plane is <u>parallel</u> to magnetic field. (دور ثان 2023)	
3	Maximum induced emf in a straight wire moving <u>parallel</u> to a magnetic field. (دور ثان 2023)	
4	Alternating current (AC) has <u>constant</u> direction and its intensity periodically change d . (تجريبي ٢٠٢٥)	
5	The ac generator is a device used for converting <u>heat</u> energy into electrical energy. (تجريبي ٢٠٢٥)	
6	The idea of <u>electric motor</u> is based on the principle of electromagnetic induction, (دور اول ٢٠٢٥)	

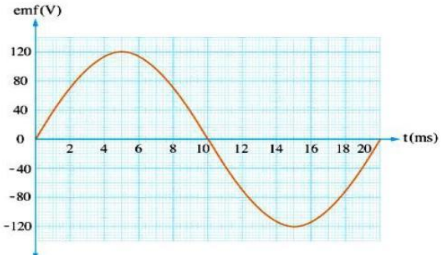
5. True or False

No.	Question	Answer
1	A straight wire moving parallel to a magnetic field produces induced electromotive force between the terminals of the wire, and if the circuit is closed induced current will flow in a direction identified by Fleming Right Hand Rule	
2	A straight wire moving in a perpendicular magnetic field produces induced electromotive force between the terminals of the wire, and if the circuit is closed induced current will flow in a direction identified by Fleming Right Hand Rule	
3	A straight wire moving in a perpendicular magnetic field produces induced electromotive force between the terminals of the wire, and if the circuit is open induced current will flow in a direction identified by Fleming Right Hand Rule	
4	AC currents are produced by generators and batteries	

5	If the wires move by an angle along the direction of the magnetic field an induced emf generated between two ends of the wire is given by: $\text{emf} = B L v / \sin \theta$. (تجريبي ٢٠٢٥)	
6	A straight wire 1 m long is moving in a uniform magnetic field of flux density 0.1 T at a uniform velocity 10 m/s. an induced potential difference of 0.5 V is generated between its terminals. Then the angle between the direction of the wire motion and the direction of the magnetic field is. 30° . (أول ٢٠٢٥)	

6. Answer the following question

No	Question	Answer
1	- A straight wire 0.5m long is moving in a uniform magnetic field of flux density 0.2T at a uniform velocity 10 m/s. An induced potential difference of 0.5 V is generated between its terminals. Find the angle between the direction of the wire motion and the direction of the magnetic field.	
2	The opposite figure shows a metallic wire (AB) of length 0.15m that is placed perpendicular to a magnetic flux of density 0.4T. When the wire is moved in the magnetic flux with a uniform velocity (v) in a certain direction, an induced emf of 0.03V is generated between its terminals which leads to the flow of an electric current from terminal (A) to terminal (B) across the wire, then the speed of the wire is and the direction of the wire velocity is (A) 1m/s, to the left of the page. (B) 1m/s, to the right of the page. (C) 0.5m/s, to the left of the page. (D) 0.5m/s, to the right of the page. 	
3	The opposite graph represents the relation between the induced electromotive force in a straight wire that moves with a uniform velocity (v) perpendicular to a magnetic flux density 0.1T and the length of that wire, so the magnitude of its velocity is (A) 1m/s (B) 2m/s (C) 0.2m/s (D) 0.1m/s Answer: (A) 1m/s 	

4	The opposite graph represents the relation between the instantaneous electromotive force (emf) produced from an AC dynamo during a complete cycle and time (t). If the face area of the dynamo's coil is $\frac{4}{\pi} \text{ m}^2$ and the number of its turns is 250 turns, then the dynamo's coil rotates in a magnetic flux density that equals (A) $3.8 \times 10^{-3} \text{ T}$ (B) $1.2 \times 10^{-3} \text{ T}$ (C) $2.6 \times 10^{-3} \text{ T}$ (D) $4.2 \times 10^{-3} \text{ T}$ Answer: (B) $1.2 \times 10^{-3} \text{ T}$	
5	State the 4 main components of the AC generator	
6	Difference between Fleming right hand and Fleming left hand rule	
7	<i>A coil of a simple AC generator consists of 100 turns, the cross-sectional area of each turn is 0.21 m^2. The coil rotates with frequency of 50 Hz (cycles/second) in a magnetic field of constant flux density $B = 10^{-3} \text{ Wb/m}^2$. What is the maximum generated induced emf? Then calculate the value of the induced emf when the angle between the direction of the linear velocity of the two longitudinal sides of the coil and the flux direction is 30°?</i>	
8	Mention the application of Fleming right hand rule.	
9	When the induced emf in straight wire moved in magnetic field equal to zero?	
10	When the induced emf in straight wire moved in magnetic field equal to maximum value?	

11	Mention the rule or method used to determine the direction of induced current in moving wire in a uniform magnetic field. (تجريبي)	
12	Mention the application of 1- electric generator. 2- Strong field magnet with concave poles in (electric generator). 3- Two slip rings in (electric generator). 4- An armature (coil) in (electric generator). 5- Two graphite brushes in (electric generator).	
13	Mention the scientific base of electric generator.	
14	A rod of copper of length 0.2 m moves with velocity 0.5m/second in a perpendicular direction to a magnetic field of density 0.8T. Calculate the emf induced in such a rod. (دور أول ٢٠٢٥)	

7. Compare between each of the following

No.	Question	Answer
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1	Compare		
	Point of comparison	direct current	alternative current
	Direction (دور أول 2023) (تجريبي)		
	Source		
	Value		
	Frequency		
	kind of produced magnetic field		
2	(تجريبي)		
	Point of comparison	The electric generator	The electric motor
	The scientific base		

Lesson (5): The electric transformer

1. Write the scientific term expressed by the following statements.

No.	Question	Answer
1	Device to step-up or step-down voltage	
2	In the transformer the ratio of output electric power to the input electric power. (دور أول / ثان ٢٠٢٣) (دور أول) (تجريبي 2024)	
3	Induced currents that circulate in closed paths due to the change in magnetic flux through a solid conductor associating with heating effect. (دور أول ٢٠٢٥) (دور أول ٢٠٢٣) (تجريبي ٢٠٢٥)	

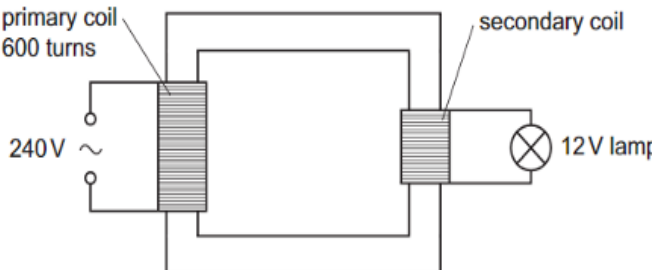
2. Mention the factors affecting each of the following (one factor only)

No.	Question	Answer
1	the induced voltage produced from the electric transformer. (تجريبي)	

3. Put circle around the correct answer

No.	Question	Choices	Answer
1	If rotational velocity of a dynamo armature is doubled, then induced e.m.f. will become	a) Half b) Two times c) 4 times d) None	
2	For a step-down transformer, the number of turns is greater in	a) primary coil b) secondary coil c) both	
3	For a step-down transformer, the volt is greater in	a) primary coil b) secondary coil c) both	
4	For a step-down transformer, the current is greater in	a) primary coil b) secondary coil c) both	
5	For a step-down transformer, the K equals to	a) =1 b) <1 c) >1 d) Not = 1	

6	In ideal transformer, the efficiency η equals to	a) $=1$ b) <1 c) >1 d) Not = 1	
7	In a not ideal transformer, the efficiency η equals to	a) $=1$ b) <1 c) >1 d) Not = 1	
8	In a step-up transformer $V_s > V_P$ implying that $I_s < I_p$. Thus a step up transformer, a) increases the voltage by increasing the current b) increases the voltage by decreasing the current c) decreases the voltage by decreasing the current d) decreases the voltage by decreasing the current		
9	In an ideal step-up transformer, which of the following statements is incorrect (دور أول 2023) (a) The current ratio of secondary to primary is smaller than one. (b) The turn's ratio of primary to secondary is smaller than one. (c) The voltage ratio of primary to secondary is greater than one. (d) The output-to-input electric power ratio is smaller than one.		
10	The ratio between the number of turns of the secondary coil to that of the primary coil in an ideal transformer is $(3/2)$, so if the output power of the transformer is (P_w) , then the input power at the primary coil equals (A) $(2/3) P_w$ (B) $1.5 P_w$ (C) P_w (D) $5 P_w$		

11	The numbers of turns of the coils in an ideal electric transformer are 400 turns and 200 turns, so if it is connected to an AC source of electromotive force 50V, then the maximum electromotive force that can be obtained is (A) 90V (B) 100V (C) 80V (D) 60V	
12	A step-down transformer is used to light a 12 V lamp from a 240 V mains supply. The lamp lights at normal brightness. The primary coil has 600 turns. How many turns are in the secondary coil? (دور أول 2025)  a) 12 b) 20 c) 30 d) 50	

4. Correct the underlined in the following statement

No.	Question	Answer
1	In transformer, core is made of soft iron to reduce <u>force opposing electric current</u>	
2	A laminated iron core is used in an electric transformer to <u>increase</u> eddy currents. (دور أول 2024)	

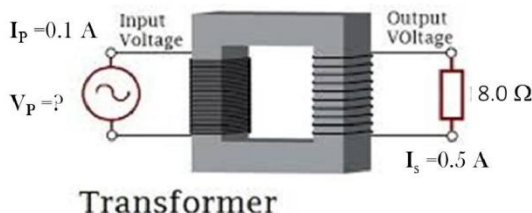
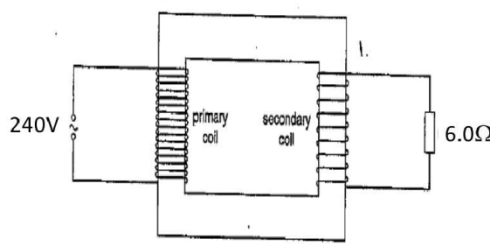
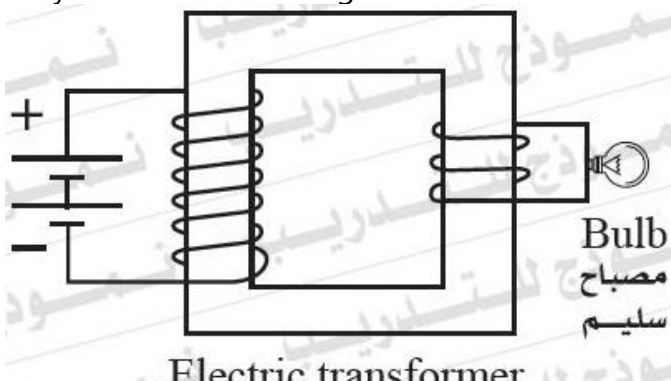
5. True or False

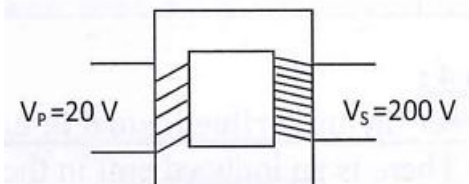
No.	Question	Answer
1	Electric transformer operates for both AC and DC currents	
2	Electric transformer operates for DC current only.	

3	Electric transformer operates for AC current only	
4	Eddy currents are associated with heating effects	
5	An AC generator can operate without a magnet	
6	An AC generator is used to convert electrical power into mechanical power	
7	An AC generator is used to convert mechanical power into electrical power	
8	DC currents have zero frequency	
9	AC currents have constant value with time	
10	DC currents produce constant magnetic field	
11	The ratio of output electric power to the input electric power in the electric transformer is called efficiency of transformer.	

6. Answer the following question

No	Question	Answer
1	An ideal step-down transformer is used to operate an electric lamp of power 24 W and potential difference 12 V using AC source of emf 240 V, if the number of turns of the secondary coil is 480 turns, calculate : (a) The electric current intensity passing through the secondary coil. (b) The number of turns of the primary coil.	

2	The secondary coil of an ideal transformer delivers an r.m.s. current of 0.5 A to a load resistor of resistance 8.0Ω. The r.m.s. current in the primary is 0.1 A. What is the r.m.s. potential difference across the primary coil? 	
3	The diagram show an iron-cored transformer assumed to be 100% efficient. The ratio of the secondary turns to the primary turns is 1:17. A 240V, a.c. supply is connected to the primary coil and a 6.0Ω resistor is connected to the secondary coil. What is the current in the primary? 	
4	An ideal step-up transformer, the ratio between the number of its turns is 7:3. Its secondary coil is connected to a device which works on a potential difference 10V. What is the value of primary coil's volt?	
5	Why does not the bulb glow? 	

6	A transformer connected to 220-volt line shows an output of 2 A at 11000 volts. The efficiency is 100%. The current drawn from the line is	
7	In a transformer the primary has 500 turns and secondary has 50 turns. 100 volts are applied to the primary coil, the voltage developed in the secondary will be	
8	The ratio between the number of turns of the secondary coil to that of the primary coil in an ideal transformer is 4/2, so if the output power of the transformer is 20 watts, then the input power at the primary coil equals to ...	
9	An ideal step-down transformer is used to operate an electric lamp, which works on a potential difference of 30 volts. If an electric source of 240 volts is used & the number of turns of the primary coil is 480 turns, calculate the number of turns of the secondary coils. (2023 دور ثان)	
10	in an ideal transformer, the secondary coil has twice as many turns as the primary. If the alternating current in the primary coil is 2A. calculate the current in the secondary coil. (تجريبي)	
11	Is this step-up or step-down transformer? Why? (تجريبي ٢٠٢٥) 	

7. Compare between each of the following

No.	Question	Answer
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1	The		
	Point of comparison	step up transformer	step down transformer
	transformer ratio value		
2	Use		
	(تجريبي ٢٠٢٥)		
	Point of comparison	ac generator	transformer
	Working principle		